CHAPTER 03: STATIC TESTING

1. Core Syllabus Concepts & In-Depth Notes

3.1 Static Testing Basics FL-3.1.1 FL-3.1.2 FL-3.1.3

> **Formal Definition:** In **static testing**, the software under test is **NOT executed**. Code, process specifications, system architecture specifications, or other work products are evaluated through manual examination (e.g., reviews) or with the help of a tool (e.g., static analysis).

Static testing assesses characteristics such as readability, completeness, correctness, testability, and consistency. Crucially, static testing can be applied for **both verification** (checking conformity to specifications) and **validation** (checking fulfillment of user and stakeholder needs in operational reality).

> **Static Testing in Agile Development:**
> Testers, developers, and business representatives (Product Owners, Business Analysts) collaborate during example mapping, user story writing, and backlog refinement. Review techniques verify that user stories meet the **Definition of Ready (DoR)** and contain clear, testable acceptance criteria.

> **Tool-Driven Static Analysis:**
> Automated tools scan code without test cases or execution. Typically integrated into **CI/CD frameworks** to enforce coding standards, evaluate code maintainability, and uncover security vulnerabilities. Spelling and grammar checkers are also static analysis tools.

3.1.1 Work Products Examinable by Static Testing FL-3.1.1 (K1)

Almost any work product generated during the software lifecycle can be examined through static testing:

Static Test Approach	Work Products That CAN Be Examined	Prerequisites & Conditions
Manual Reviews	• Requirements specifications & User stories • Architecture and system design documents • Source code and database schemas • Test plans, test cases, test charters, test procedures • Product backlog items, contracts, project plans, models	Any work product that can be **read and understood by human beings**.
Automated Static Analysis	• Source code and scripts • Data models and software architectures • Formal text specifications with standardized grammar	Requires a **formal structure or syntax** against which automated rules can check.

EXAM ALERT — What Work Products are NOT Appropriate for Static Testing?
Work products that are *difficult to interpret by human beings AND should not be analyzed by tools*. A prime example is **3rd-party executable binary code** (commercial off-the-shelf compiled software) due to legal, licensing, or intellectual property restrictions.

3.1.2 Value of Static Testing FL-3.1.2 (K2)

Static testing provides exceptional organizational and economic value throughout the SDLC:

- **Earliest Defect Detection (Shift Left):** Finds defects in early requirements and design phases before implementation starts, embodying Principle 3 (Early testing saves time and money).
- **Detects Defects Undetectable by Dynamic Testing:**
  - *Unreachable / dead code:* Code blocks that can never be executed during runtime.
  - *Architectural / design pattern violations:* Inappropriate design patterns or poor modularity.
  - *Defects in non-executable artifacts:* Ambiguities, contradictions, and omissions in requirements specifications, contracts, and test plans.
- **Reduced Total Cost of Quality:** Although reviews require upfront time and effort, total project costs are significantly lower than without reviews because downstream rework costs drop dramatically.
- **Shared Understanding & Communication:** Bringing diverse stakeholders together early fosters mutual alignment, shared vision, and eliminates requirements misunderstandings.

3.1.3 Differences between Static Testing and Dynamic Testing FL-3.1.3 (K2)

Static testing and dynamic testing are complementary disciplines with similar goals (detecting defects and evaluating quality), but distinct mechanisms:

Dimension	Static Testing (Reviews & Static Analysis)	Dynamic Testing (Execution Based)
Execution of Code	Software is **NOT executed**.	Software **MUST be executed** with inputs.
Defect Identification	Finds defects **directly** in the work product.	Causes **failures**; defects are found via debugging.
Applicability	Applies to executable AND non-executable work products.	Applies **only to executable work products**.
Rare Paths Coverage	Easily detects defects on rarely executed or hard-to-reach paths.	Difficult to trigger obscure edge-case execution paths.
Quality Focus	Measures non-runtime characteristics (e.g., **maintainability**, consistency).	Measures runtime characteristics (e.g., **performance efficiency**, reliability).

Typical Defects Easier & Cheaper to Find Through Static Testing:

Requirements & Design Flaws:

- Inconsistencies, ambiguities, and contradictions.
- Omissions, inaccuracies, and unnecessary duplications.
- Inefficient database structures and poor modularization.
- Gaps or inaccuracies in test basis coverage.

Code & Interface Defects:

- Undefined or undeclared variables.
- Unreachable, dead, or duplicated code.
- Excessive code complexity and standard violations.
- Mismatched interface parameters (number, type, order).
- Security vulnerabilities (e.g., buffer overflows).

1.2 Feedback and Review Process

3.2.1 Benefits of Early and Frequent Stakeholder Feedback FL-3.2.1 (K1)

Continuous stakeholder involvement prevents divergence from customer expectations. If stakeholder engagement is minimal, the delivered product may deviate from business goals, resulting in expensive rework, missed deadlines, blame games, or project cancellation. Early

feedback clarifies requirements changes early and ensures teams focus on features delivering the highest value and mitigating critical risks.

3.2.2 Review Process Activities (ISO/IEC 20246) FL-3.2.2 (K2)

The international standard **ISO/IEC 20246** defines a generic review framework tailored to project formality. For large work products, the review process is invoked **multiple times** across small chunks:

1. Planning
2. Review Initiation
3. Individual Review
4. Communication & Analysis
5. Fixing & Reporting

Activity	Core Tasks & Responsibilities	Primary Purpose
1. Planning	Defines review scope: purpose, work product to review, quality characteristics to evaluate, areas to focus on, exit criteria, supporting documents (standards), effort, and timeframes.	Establish review boundaries, criteria, and logistics.
2. Review Initiation	Ensures everyone and everything is prepared to start: distributes work product under review, verifies access, clarifies roles/responsibilities, and provides supporting review materials.	Ensure all participants are ready and equipped.
3. Individual Review	Each reviewer individually assesses the work product. Identifies anomalies, recommendations, and questions using techniques (checklists, scenarios). Reviewers log all identified items.	Uncover anomalies independently without groupthink.
4. Communication & Analysis	Anomalies are discussed (anomalies are not necessarily defects!). Decisions are made on status, ownership, and required actions. Evaluates product quality level and determines if follow-up review is needed.	Analyze anomalies, assign fixes, determine quality status.
5. Fixing & Reporting	Defect reports are created for confirmed defects so corrective actions are followed up. Author fixes work product. Work product accepted once exit criteria are met. Review results are reported.	Remediate defects, confirm exit criteria, report outcomes.

EXAM ALERT — Anomalies vs. Defects in Reviews: An *anomaly* is any potential issue, question, or deviation noted during an individual review. During the *Communication and Analysis* meeting, anomalies are evaluated; only confirmed flaws become official *defects* requiring corrective action.

3.2.3 Roles and Responsibilities in Reviews FL-3.2.3 (K1)

Reviews involve multiple stakeholders, each assuming specific responsibilities according to **ISO/IEC 20246**:

Principal Role	Core Responsibilities & Syllabus Definition	Exam Key Takeaways
Manager	Decides **what is to be reviewed** and provides resources, such as staff, funding, and time for the review.	Allocates resources; does NOT conduct the technical meeting.
Author	Creates and fixes the work product under review.	Resolves defects; *in inspections, CANNOT act as Review Leader or Scribe!*
Moderator (Facilitator)	Ensures the effective running of review meetings, including mediation, time management, and maintaining a **safe review environment** in which everyone can speak freely.	Leads the meeting discussions; neutral mediator.
Scribe (Recorder)	Collates anomalies from reviewers and records review information, including decisions and new anomalies identified during the review meeting.	Documents review outcomes; keeps the official review record.
Reviewer	Performs reviews by examining the work product to identify anomalies. May be a team member, Subject Matter Expert (SME), or stakeholder.	Applies critical thinking to detect potential defects.
Review Leader	Takes **overall responsibility** for the review: decides who participates, and organizes when and where the review will take place.	Administrative and organizational owner of the review.

Inspection Independence Rule (CRITICAL):

In formal **Inspections**, the **Author CANNOT** act as the **Review Leader** or **Scribe**. This preserves objectivity, prevents defensive bias, and ensures impartial documentation of defect data.

Role Sharing & Flexibility:

In less formal reviews, roles may be combined (e.g., the Review Leader may also act as Moderator or Reviewer). However, separation of duties increases objectivity in formal audits.

3.2.4 The Four Review Types FL-3.2.4 (K2)

Reviews range from informal pair check-ins to rigorous formal inspections. The required level of formality depends on SDLC maturity, product criticality, regulatory standards, and audit requirements:

Review Type	Formality & Process	Meeting Leadership	Primary Objectives	Key Distinguishing Traits
Informal Review	Low formality; no defined formal process.	Typically informal (e.g., peer, pair).	Detecting anomalies.	No formal documented output required; inexpensive and agile.
Walkthrough	Low to medium formality.	**Led by the Author** of the work product.	Evaluate quality, build confidence, educate reviewers, gain consensus, generate new ideas, motivate author, detect anomalies.	Author explains work product step-by-step; individual prep is **optional**.
Technical Review	Medium to high formality.	Led by a trained **Moderator**.	Gain consensus and make decisions regarding a technical problem; detect anomalies; evaluate quality.	Conducted by **technically qualified reviewers**; individual review preparation is expected.
Inspection	**Most formal** review type. Follows complete ISO 20246 process.	Led by a trained **Moderator / Review Leader**.	Find the **maximum number of anomalies**; evaluate quality; build confidence; collect process improvement metrics.	Uses entry/exit criteria, formal checklists and rules; metrics collected; *Author cannot be leader or scribe!*

Choosing the Right Review Type (Context-Driven):

The same work product can undergo multiple review types over its lifecycle (e.g., an informal peer review first, followed by a formal inspection prior to regulatory submission). Selection depends on: project risks, available budget, regulatory/legal audit needs, and organizational maturity.

Individual Review Techniques (ISO/IEC 20246):

During individual review, reviewers apply specific techniques:

- **Ad hoc:** Unstructured reading with little or no guidance.
- **Checklist-based:** Systematically checking against predefined defect types.
- **Scenarios & Dry Runs:** Simulating execution using realistic use cases.
- **Role-based:** Evaluating from specific stakeholder roles (e.g. operator).
- **Perspective-based Reading (PBR):** Adopting distinct viewpoints (user, designer, tester) and drafting artifacts (e.g. tests). *Empirically yields highest defect detection.*

3.2.5 Success Factors for Reviews FL-3.2.5 (K1)

A successful review program depends on both organizational culture and meticulous procedural execution. The syllabus identifies ten critical success factors:

1. Clear Objectives & Exit Criteria

Define unambiguous goals and measurable exit criteria for every review. *Crucial: Evaluation of participants must NEVER be an objective!*

2. Right Review Type for Context

Select the appropriate review type to match the work product complexity, risks, participant skill set, and regulatory needs.

3. Review in Small Chunks

Break large work products into small chunks so reviewers maintain high concentration and vigilance during individual reviews and meetings.

4. Constructive Feedback Loops

Provide actionable, objective feedback to authors and stakeholders so work products and development processes are continuously improved.

5. Adequate Preparation Time

Participants must be granted sufficient dedicated time in their project schedule to thoroughly conduct individual reviews prior to meetings.

6. Executive Management Support

Management must actively endorse the review process, allocating realistic budget, scheduling allowance, and resourcing.

7. Embedded Learning Culture

Embed reviews into organizational culture as collaborative opportunities for team learning and process improvement rather than fault-finding.

8. Participant Training

Ensure all reviewers, moderators, and authors receive proper training on how to fulfill their specific roles effectively.

9. Effective Meeting Facilitation & Psychological Safety:

Review meetings must be mediated by a skilled moderator who manages time, mediates conflicts, and establishes a safe psychological environment where all participants speak openly without fear of criticism.

CRITICAL EXAM TRAP: Participant Evaluation is Forbidden!

Reviews are exclusively designed to evaluate the **quality of the work product**. Using review metrics (e.g. defects found) to appraise or evaluate participant performance poisons team trust, breeds defensive behaviors, and destroys the review culture.

2. High-Yield Exam Flashpoints & Key Distinctions

TOP 10 ISTQB CTFL v4.0.1 CHAPTER 3 EXAM TRAPS & DISTRACTORS:

1. **"Static testing cannot perform validation"** → **FALSE!** Static testing supports *both verification and validation* (e.g., validating user stories against user operational needs).
2. **"Static analysis requires test cases"** → **FALSE!** Static analysis examines code and models using automated rule-sets without executing code or designing test cases.
3. **"Individual preparation is mandatory in walkthroughs"** → **FALSE!** While reviewers *might* review individually beforehand, individual preparation is *optional* in walkthroughs.
4. **"Anomalies found in reviews are always defects"** → **FALSE!** Anomalies are questions, ambiguities, or deviations; only after discussion in the meeting are confirmed issues logged as defects.
5. **"The author can lead an inspection"** → **FALSE!** In formal inspections, the author *cannot* act as the Review Leader or Scribe. Walkthroughs, however, are author-led!
6. **"Review metrics should be used to evaluate employee performance"** → **FALSE!** Participant evaluation must *never* be an objective; reviews evaluate product quality only.
7. **"Static testing replaces dynamic testing"** → **FALSE!** Static and dynamic testing are complementary; each finds distinct defect classes (e.g., dynamic finds runtime performance issues).
8. **"Inspections have no formal entry or exit criteria"** → **FALSE!** Inspections are the most formal review type and strictly enforce documented entry and exit criteria.
9. **"Third-party compiled binary code is ideal for static analysis"** → **FALSE!** Binary code lacking source access is unsuitable for static analysis due to legal and interpretability limits.
10. **"Technical reviews are led by the author"** → **FALSE!** Technical reviews are led by a trained *Moderator* and performed by technically qualified peers.

Rapid Comparison Matrix (Core Review Types & Roles)

Review Type	Leader / Facilitator	Individual Preparation	Core Objective
Informal Review	Peer / Author	Not required	Detecting anomalies quickly without formal documentation.
Walkthrough	**Author**	Optional	Educate reviewers, build consensus, motivate author, detect anomalies.
Technical Review	**Moderator**	Expected / Recommended	Gain consensus and resolve technical architectural problems.
Inspection	**Moderator / Leader**	**Mandatory**	Find maximum number of anomalies, enforce rules, collect metrics.

High-Yield Memory Mnemonics for Revision:

- **5 Review Steps:** *Plan → Initiate → Individual Review → Communicate & Analyze → Fix & Report.*
- **4 Review Types:** *Informal (No rules) • Walkthrough (Author-led) • Technical (Tech consensus) • Inspection (Max formal).*
- **Key Role Rules:** *Manager (Resources) • Moderator (Meeting safe/on time) • Scribe (Records) • Leader (Overall review owner).*

3. Complete Chapter 3 Keyword Lexicon (All 10 Official Terms)

Anomaly Any condition that deviates from expectation based on requirements specifications, design documents, user documents, standards, or experience.

Dynamic Testing Testing that involves the execution of the software of a component or system.

Formal Review A form of review that follows a defined process with a documented output, formal roles, and defined entry/exit criteria (e.g., an inspection).

Informal Review A review that does not follow a defined process and does not require a formal documented output.

Inspection The most formal type of review, based on visual examination of work products to detect anomalies, using rules, checklists, metrics, and entry/exit criteria.

Review An evaluation of a work product to detect anomalies, improve quality, and ensure conformity with standards and requirements.

Static Analysis The process of evaluating a component or system without executing the software, based on its form, structure, content, or security properties.

Static Testing Testing a work product without code being executed, comprising reviews and static analysis.

Technical Review A formal review by technically qualified reviewers, led by a moderator, to gain consensus on a technical problem and detect anomalies.

Walkthrough A review led by the author of the work product to evaluate quality, build confidence, educate reviewers, and detect anomalies.

Key Terminology Relationship:

Static Testing is the umbrella discipline comprising both **Human Reviews** (Informal, Walkthrough, Technical Review, Inspection) and **Static Analysis** (automated linting, complexity checks, security scanning).

STUDENT NOTES & EXAM FLASHCARD REMINDERS (Handwritten Study Space)

4. Realistic ISTQB Practice Exam Questions & Detailed Rationales (Part 1)

Question 1 [K1 - FL-3.1.1]: Which of the following work products is LEAST APPROPRIATE for examination using static testing?

A) A software architecture diagram detailing microservice dependencies.

B) A third-party proprietary compiled executable binary where source code is legally inaccessible.

C) A product backlog containing user stories and acceptance criteria.

D) An automated test execution script written in Python.

Correct Answer: B
Rationale: Section 3.1.1 explicitly states that work products not appropriate for static testing are those that are difficult to interpret by humans and should not be analyzed by tools, such as third-party executable code due to legal restrictions. Architecture, user stories, and test scripts are all prime candidates for static testing.

Question 2 [K2 - FL-3.1.2]: Which of the following defects is TYPICALLY detected by static testing but CANNOT be detected by dynamic testing?

A) Excessive memory leakage occurring after 48 hours of continuous server load.

B) An unreachable block of dead source code that can never be executed.

C) Slow database query response times during peak holiday transaction traffic.

D) Inaccurate calculations triggered by unexpected floating-point precision errors during runtime.

Correct Answer: B
Rationale: Dynamic testing requires code execution; therefore, code that is unreachable (dead code) can never be executed or detected dynamically. Static analysis identifies dead code directly without execution (Section 3.1.2). Memory leaks, response times, and calculation errors during execution are dynamic phenomena.

Question 3 [K2 - FL-3.1.3]: How does static testing fundamentally differ from dynamic testing regarding defect detection?

A) Static testing triggers failures from which defects are diagnosed, whereas dynamic testing identifies defects directly.

B) Static testing finds defects directly in work products, whereas dynamic testing causes runtime failures from which defects are determined.

C) Static testing can only be applied to executable code, whereas dynamic testing applies to documentation.

D) Static testing measures performance efficiency, whereas dynamic testing evaluates code maintainability.

Correct Answer: B
Rationale: Section 3.1.3 explains that static testing detects defects directly in the examined artifact, whereas dynamic testing triggers observable runtime failures, requiring subsequent debugging to isolate the defect.

Question 4 [K1 - FL-3.2.1]: What is a primary benefit of gathering early and frequent stakeholder feedback during the SDLC?

A) It guarantees that the system will achieve 100% statement and branch code coverage.

B) It eliminates the need to perform component and system integration testing.

C) It prevents misunderstandings regarding requirements and ensures changes are implemented before expensive rework is required.

D) It ensures that the author of the code acts as both moderator and scribe during formal reviews.

Correct Answer: C
Rationale: Frequent feedback prevents misunderstandings about requirements, ensures changes are understood earlier, and avoids costly rework and missed deadlines (Section 3.2.1).

4. Realistic ISTQB Practice Exam Questions & Detailed Rationales (Part 2)

Question 5 [K2 - FL-3.2.2]: During which activity of the review process (ISO/IEC 20246) are anomalies discussed, their status decided, and ownership assigned?

A) Review initiation

B) Individual review

C) Communication and analysis

D) Planning

Correct Answer: C
Rationale: Section 3.2.2 explicitly defines that during *Communication and analysis*, participants analyze and discuss logged anomalies, make decisions on their status and ownership, and decide on necessary corrective actions.

Question 6 [K1 - FL-3.2.3]: In an inspection, which two roles MUST NOT be performed by the author of the work product?

A) Reviewer and Manager

B) Review Leader and Scribe

C) Moderator and Reviewer

D) Scribe and Reviewer

Correct Answer: B
Rationale: Section 3.2.4 strictly stipulates: "In inspections, the author cannot act as the review leader or scribe." This prevents conflict of interest and ensures impartial documentation of defects.

Question 7 [K2 - FL-3.2.4]: Which review type is led by the AUTHOR of the work product and has multiple objectives including educating reviewers, building consensus, and gathering new ideas?

A) Walkthrough

B) Inspection

C) Technical Review

D) Informal Review

Correct Answer: A
Rationale: Section 3.2.4 defines a *Walkthrough* as being led by the author, serving objectives such as educating reviewers, building consensus, generating new ideas, and detecting anomalies.

Question 8 [K1 - FL-3.2.5]: Which of the following is an ANTI-PATTERN that directly undermines the success of a review program?

A) Conducting reviews on small chunks to avoid reviewer fatigue.

B) Using defect metrics gathered from reviews to evaluate the performance of participating authors.

C) Providing adequate preparation time for participants to perform individual reviews.

D) Having a trained moderator facilitate review meetings to maintain a safe environment.

Correct Answer: B
Rationale: Section 3.2.5 explicitly emphasizes: "Evaluation of participants should never be an objective." Using reviews to appraise employee performance creates fear and ruins the review culture.